



Sterile Corridor Design as a Resilience Problem

An Analytical Report Generated by the AI Research Council

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Abstract

Sterile corridor design is treated as a code-compliance problem when it should be treated as a throughput and resilience problem. The difference between a sterile circulation system that scales and one that becomes the terminal's permanent bottleneck is set 10+ years before passengers ever walk through it — in decisions about width, redundancy, vertical circulation, and what happens when one segment fails. MWAA needs a framework for evaluating sterile circulation not against TSA minimums but against the operational reality of recovery from disruption. The report examines this claim across the following: A sophisticated airport planner cannot dismiss the piece in the first 500 words — the framing has to land as obviously correct and obviously underused; The 4–6 design variables identified are specific enough to apply to a real corridor on a plan set, not abstract enough to apply to everything; Width-to-throughput claims trace to empirical sources (measured pedestrian flow studies, post-occupancy evaluations, observed peak-hour conditions), not to LOS letter grades or designer rules of thumb; The IAD application is concrete enough that a MWAA planning staffer reads it and says "this changes how we should evaluate the next Tier 2 segment"; The counter-case — that sterile corridors are already over-designed, that the real bottleneck is checkpoints or gates, that operational fixes are cheaper than capital ones — is presented strongly enough that the reader feels its force before the piece answers it.

How This Report Was Produced

This report is the product of the Transform Airports AI Research Council — a multi-agent analytical system running on Anthropic's Claude models. Rather than a single model writing in one pass, the Council convenes a panel of specialized research agents, subjects their work to adversarial review, and verifies every claim before publication. The process runs in four stages, with human review at two checkpoints.

Stage 1 — Independent research. Each research agent below produced an evidence brief without seeing the others' work, so the analysis draws on genuinely independent lines of evidence rather than a single line of reasoning.

Stage 2 — Synthesis and adversarial review. A Strategist synthesized the briefs into an argument; a Red Team attacked it across two rounds; the Strategist revised after each.

Stage 3 — Edit and fact-check. An Editor tightened the prose. A Fact-checker verified every number against the research briefs, removing or flagging what it could not source.

Stage 4 — Production. The verified draft was formatted into this document.

Research agents seated for this report

- Airline Commercial Strategist
- Airport CEO
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Process agents (on every report)

- **Strategist** — synthesized the research briefs into a single argumentative draft.
- **Red Team** — attacked each draft to surface weak claims and logical gaps.
- **Editor** — tightened the prose and removed filler without adding new content.
- **Fact-checker** — verified every numerical and attributed claim against the briefs.

Each agent is governed by a versioned specification — its analytical mandate, evidentiary standards, source discipline, and argumentative posture. The complete technical definitions for every agent on the Council are published and openly available at:

<https://github.com/TransformAirports/ai-council/tree/main/.claude/agents>

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I. The Cleaning Crew at 6:45 a.m.

At 6:45 a.m. on a Tuesday in February, the Concourse C escalator bank at Dulles is doing the work it does most mornings — moving roughly two hundred passengers from an AeroTrain that arrived at 6:39 up to the sterile concourse level, where four United banks are converging on a forty-minute departure window. One of the two escalators is running rough. Facilities flagged it on the 0500 walk; it has not tripped out yet. A cleaning crew is at the south end of the corridor, fifteen minutes past their scheduled clearance, because somebody spilled coffee at 5:15 and the second pass took longer than the contract assumes. The TSA checkpoint upstream is at three of four lanes. One officer called out. This is normal for a Tuesday in February, the kind of normal an operator with twenty-five years in the seat knows by feel.

Nothing is broken. Everything is marginal.

The corridor between the AeroTrain landing and the gate hold rooms at C18 through C28 is, on a plan set, a Level of Service C facility. The Highway Capacity Manual says LOS C accommodates roughly 600 pedestrians per hour per foot of effective width, and the corridor was sized to that standard when HOK drew it in the mid-1990s. What the plan set did not contain is that the dominant flow this morning is bidirectional — the departing bank from the gates running headlong into the arriving bank off the train — and a 2018 PLOS One synthesis of pedestrian flow studies found that bidirectional capacity clusters around 1.5 persons per meter per second versus roughly 2.0 for unidirectional flow, a 25 percent reduction. The plan set also did not contain the cleaning cart in the middle of the corridor, which a 2019 *Physica A* controlled experiment found concentrates density 16 percent higher than non-parallel obstacle layouts at the same volume. It did not contain that one escalator is running rough.

Some of what is going wrong here is contractual, not dimensional. The cleaning window at IAD's Concourse C runs from 0230 to 0530 and routinely runs over to 0545 or 0600, because the facilities contract was negotiated against an operating schedule that predates the current United hub bank structure and the facilities metric is task completion, not corridor performance. Fix that contract and the 0700 bank gets a little easier on most days. What the contract fix cannot do — and this is the load-bearing distinction the rest of this piece will earn — is restore an escalator that is out of service, summon a second one that was never built, or rewrite the geometry of a corridor whose width was set when the column grid went in.

The corridor is not failing. It is operating in a regime that the design standard does not describe and the post-occupancy literature has not measured. No airport reviewed for

this run has published a peer-reviewed comparison of as-built Level of Service predictions against measured peak-hour throughput five or more years after opening. The industry is designing to a model it has never validated, against a 1971 dataset that the Airport Cooperative Research Program itself, in Problem Statement 24-1058 (FY2024), acknowledges does "not account for demographic trends or increased prevalence of roll-aboard luggage."

Here is the thesis, in three sentences.

Sterile corridor design at major US hub airports is governed by code minimums and a fifty-year-old pedestrian flow framework that was never built for what these corridors are now asked to do. The specific failure modes worth worrying about are not slow congestion at peak but binary vertical-circulation failures at APM handoffs, whose 1990s installed base is now reaching end-of-life simultaneously across the midfield generation of US hubs. MWAA, with the Tier 2 program in active construction and a \$22 billion revitalization concept under DOT review with no confirmed federal funding mechanism, has perhaps eighteen months before the next generation of sterile circulation at IAD locks in for thirty years.

The rest of this piece argues that this is true, that the cheaper alternatives the contrarian case names — operational scheduling, maintenance SLAs, checkpoint capital — are necessary but insufficient against the failure modes the AeroTrain handoff geometry generates, and that the path-dependence argument is more bounded than it first appears but still binding for reasons rooted in MWAA's particular capital and political position. The corridor is not the only problem at IAD. It is the only problem that has no operational substitute when the asset itself is gone.

II. Executive Summary

The Level of Service framework that governs sterile corridor design at midfield airports assumes a flow it does not receive — pulsed, bidirectional, luggage-loaded, with binary vertical-circulation failure modes at the APM handoff. The 1990s midfield generation built to that framework is reaching end-of-life on its first vertical circulation cycle at exactly the moment the post-pandemic traffic peak and the post-IIJA capital cycle converge. Operational interventions move the middle of the failure distribution; only redundant infrastructure compresses the tail where the asset itself is gone. MWAA's specific window is open now. If the Tier 2 program closes without resetting handoff geometry, the next opportunity is 2040-plus.

The eight specific claims that argument rests on:

1. **A corridor rated LOS C on a design drawing is operating at functional LOS D during the dominant directional surge of an AeroTrain discharge.** Maximum bidirectional flow is 25 percent below the unidirectional flow on which LOS tables were calibrated, per the 2018 PLOS One synthesis. Add the cleaning-crew obstacle penalty (16 percent density concentration, per the 2019 *Physica A* study)

and the luggage-laden walking-speed reduction (3 to 8 percent, per Seth Young's 1999 TRB study), and the design LOS becomes a fictional grade.

2. **The AeroTrain delivers pulses, and the LOS framework does not model pulses.** Three-car trains with a design capacity of 216 passengers (72 per car) arrive on roughly two-minute peak headways. Operational loads run higher — the COO's chair memory puts a peak discharge at closer to 280 passengers when the system is on reduced equipment and trains accumulate. The divergence between nameplate and observed itself supports the thesis: the design model contains neither pulse arrival nor the operational stress that drives observed load above nameplate.
3. **The empirical foundation of current design standards is a 1970s dataset that ACRP itself has formally acknowledged is no longer valid.** ACRP Problem Statement 24-1058 (FY2024): "the basic research underlying existing formulas was created in the 1970s" and "does not account for demographic trends or increased prevalence of roll-aboard luggage." FAA AC 150/5360-13A continues to reference the unflagged Fruin-derived tables as the design starting point.
4. **No airport in the comparator set has published a post-occupancy evaluation comparing pre-opening simulation predictions to measured corridor throughput.** A 2025 ACM conference paper on airport simulation practice found no such validation references in its literature search. The industry uses MassMotion, AnyLogic, and SimWalk to certify designs and then never reopens the models.
5. **Denver International is spending \$300 to \$700 million to add a pedestrian alternative to its underground train, thirty-one years after the original decision to build without redundancy.** A 99 percent uptime APM produces three to four full failures per year. In those moments, the only access route to the concourses fails completely. The 1995 design choice, made under the same operations-substitutes-for-capital theory the contrarian case offers today, is being corrected at almost three quarters of a billion dollars in 2028 money — exclusive of three decades of distributed operational cost that never appeared in any capital line.
6. **In-place sterile renovation at occupied concourses is possible, and DCA Project Journey proves it. The price is severe phasing constraint.** Construction windows of 6.5 hours per night at DCA, shrinking to roughly 2.5 hours per night at IAD's AeroTrain station locations because the train operates earlier and later than the gates it serves. The chief-engineer brief, flagged as practitioner observation rather than published figure, puts in-place sterile-side construction productivity at roughly half to two-thirds of greenfield equivalents. The question is not whether renovation is possible. It is whether the geometry can be reset within the structural envelope the column grid permits, and whether the financial multiplier on mid-life work makes the right answer a generational commitment instead of a recurring one.
7. **The political economy is structurally hostile to the investment.** The PFC cap of \$4.50, unchanged since 2000 and reaffirmed by Congress in P.L. 118-63 (May 2024), has lost roughly 55 percent of its purchasing power against the ENR Construction Cost Index — the right baseline for a capital discussion, and the one the historian brief flagged as more honest than the 40 percent CPI figure that often

appears in advocacy materials. United Airlines holds something on the order of 50 to 65 percent of IAD enplanements depending on which dataset and metric you read (the CEO brief reads 50 to 55 percent of seats; the airline-strategist brief reads 60 to 65 percent of enplanements; both should be verified against current MWWA and T-100 data). At any of those values, United triggers Majority-in-Interest review on material common-use capital, and sterile corridor capital produces no revenue, no ribbon-cutting, no rating-agency line item, and no metric in carrier dashboards.

8. **The IAD midfield concourses are at the inflection point.** A/B opened 1997; C/D 2000; the AeroTrain was retrofit onto an already-built sterile envelope in 2010. The 1990s vertical circulation, designed for 20-year service life and closer to 15 to 20 under actual airport use, is at or past end-of-life. The next capital cycle — roughly 2027 to 2035 by MWWA's CIP framing — is the window. If it closes without resetting corridor geometry and vertical circulation redundancy at the AeroTrain handoffs, the next opportunity is 2040-plus.

III. The Argument

The Pulse: Watching One Arrive

It is 6:39. The northbound AeroTrain decelerates into the C station and the platform doors release a wash of passengers — most coming up from the main terminal off arriving banks between 6:10 and 6:25, some international connectors who cleared CBP an hour ago and are walking faster than the local-origin travelers because they have been moving since 4 a.m. Eastern. The two escalators carry them up. One is running rough but still moving. The platform clears in roughly the time it takes to clear it. Two minutes later, the next train arrives behind it.

This is what the AeroTrain does at peak. It does not deliver a flow. It delivers a pulse, and the corridor downstream — through the gate hold areas at C18 to C28 — receives the pulse against the counter-flow of departing passengers walking the other way. The Highway Capacity Manual's pedestrian framework, derived from John Fruin's 1971 work, was built for transit stations and urban sidewalks where arrival is treated as approximately Poisson — distributed, irregular, but not pulsed. Fruin measured pedestrians on city streets and in subway concourses, not airports, and his Level of Service thresholds were never intended for surge-load scenarios. Their adoption as the airport corridor design standard was a planning-community decision, not a Fruin recommendation.

The stadium industry, facing a load profile aviation has not chosen to recognize as comparable, developed a different vocabulary. The UK Green Guide, derived from Lord Justice Taylor's report on the 1989 Hillsborough disaster, specifies maximum flow rates of 82 persons per meter per minute on flat surfaces — roughly 2.5 to 3 times the comfortable LOS C flow rates of Fruin-derived airport standards. Hillsborough killed 97 people because the wrong shape of crowd met the wrong shape of corridor at the wrong rate. It produced the most rigorous codified rethink of crowd circulation under surge in

any venue category. Airports have not had their Hillsborough. The Taylor framework is sitting there, available to be analogized, and the AeroTrain-to-concourse handoff geometry is a stadium concourse problem wearing an airport coat. The question the slacker brief asks — does MWAA want to have its Hillsborough proactively, or reactively — is the right one to put to the board. The number 82 belongs in the planning vocabulary the same way LOS C does, and for the same reason: it is what the corridor must do in the worst minute, not what it must feel like in the average one.

Here is the arithmetic. The calculation is analyst-constructed from brief-cited components; the operational point survives the caveat. A 20-foot (6-meter) approach corridor at LOS C carries roughly 200 persons per minute in unidirectional flow under the HCM tables. Apply the 25 percent bidirectional penalty from the PLOS One synthesis and effective capacity falls to about 150 persons per minute. A three-car AeroTrain with a 216-passenger design capacity, discharged through two escalators in the seconds it takes to clear a fully-loaded platform, puts a load on the corridor at or above its effective bidirectional capacity. The buffer is gone before the next train arrives two minutes later. The 280-passenger COO observation, under reduced-equipment operations, puts the load above effective capacity by roughly 50 percent. No published study of the IAD AeroTrain handoff has measured the actual pulse-discharge duration, and the operations-analyst brief is explicit that field verification is required. The corridor is not buffer-less. But the design framework assumed a flow rather than a pulse, no one has measured the pulse, and the order-of-magnitude arithmetic says the buffer is small enough that measurement should not be optional.

The 2019 *Physica A* paper compounds the picture counterintuitively. Wider approach corridors upstream of a fixed bottleneck can funnel higher-density crowds into the bottleneck transition. Width without redundancy is the wrong fix. Corridor width matters, but the binding constraint is the vertical transition at the APM handoff, and adding width upstream without solving the vertical transition can move the failure point an arm's length closer to the escalator without compressing it.

The 1970s Dataset That ACRP Has Disavowed

The most important sentence in the regulatory record on this question was published by ACRP itself in 2024.

Problem Statement 24-1058, in the FY2024 cycle, states: "the basic research underlying existing formulas was created in the 1970s, with some efforts to customize these to specific airport environments, though these are over twenty years old and do not account for demographic trends or increased prevalence of roll-aboard luggage." This is the industry's own research program acknowledging that the design tools currently in use are calibrated to a passenger profile that no longer exists.

The carry-on baggage profile is the cleanest example. Modern airport passengers carry significantly more carry-on luggage than passengers of the 1970s — a direct response to checked-bag fees that did not exist in Fruin's sample. The contrarian brief cites IATA's 2023 Baggage Report on North American carry-on rates rising from 0.7 to 0.8 per

passenger to roughly 1.3 in 2022; the figure should be verified against the underlying IATA report, but the directional claim is uncontested in the industry literature. Fruin's effective width calculations assumed a pedestrian body ellipse of approximately 0.5 by 0.6 meters. A passenger with a roller bag trailing behind occupies a longitudinal footprint closer to 1.5 to 2.0 meters.

FAA AC 150/5360-13A (2018), the advisory circular that governs domestic terminal planning, does not flag the gap. It continues to reference pedestrian flow capacity in persons per unit width per minute, derived from Fruin, without noting that the underlying data predates roll-aboard luggage proliferation. There is no enforcement mechanism either way; the AC explicitly describes itself as "a reference document and starting point" — not a regulatory mandate. The FAA cannot cite an airport for having a 22-foot corridor instead of a 30-foot one. Its lever is funding eligibility, not citation authority. The result in practice is that designers reach for the AC's referenced figures as a ceiling, not a floor, and the column grid the building inherited is the actual binding constraint.

The Column Grid and the Geometry the Building Will Not Surrender

The chief engineering reality at any 1990s midfield concourse is that the corridor's usable width was determined not by a pedestrian flow analysis but by what was left after the structural bay accommodated gate hold-rooms, MEP chase space, and mechanical interstitial. The IAD midfield concourses, drawn by HOK and built in phases between 1997 and 2004, inherited a structural grid sized for the gate program of late-1990s United hub operations. Widening a corridor after construction requires either relocating columns — structurally prohibitive without temporary shoring of the floors above — or cantilevering into the existing bay, which eats into MEP chase space along the corridor ceiling. The choice between narrower corridors and undersized mechanical is not a design preference. It is a structural inheritance that persists for the life of the building.

This is the constraint the renovation argument runs into. Renovation is real. The question is what renovation can do.

Three things sit fixed for the life of the concourse inside the inherited envelope. Column spacing fixes the maximum clear span and therefore the upper bound on how far a corridor can widen between columns without a structural transfer. The MEP chase runs along the corridor ceiling — relocate it and the cost cascades into the active mechanical systems serving the gates above. Third, and surfaced most sharply in the chief-engineer brief, is the moving-walkway pallet-width and exit-throat geometry. The 28- and 32-inch pallet-width walkways installed in 1990s midfields were sized to the volumes of that decade. The exit throat — the point at which a passenger steps off the moving belt into the static corridor — is governed by the balustrade termination angle and the clear width of the receiving corridor. At IAD's midfield connectors, the exit-throat effective width is narrower than the walkway pallet width, because the balustrade overhangs and a column sits at the corridor intersection. The result is exactly the behavior visible at morning bank peaks: passengers accumulating at the walkway exit, creating a backflow onto the moving surface itself, which is a fall-hazard condition and a throughput loss that no PA

announcement or signage corrects. The exit throat is a designed-in constraint that no maintenance SLA, no cleaning-window adjustment, and no checkpoint expansion will touch.

What renovation can change is bounded by what the original column grid and chase layout permit. The right time to influence the geometry of the next-generation midfield envelope is the design-development phase of the work being committed at IAD right now. The Tier 2 East concourse opens September 2026. The geometry that opens that day is the geometry that the next thirty years of midfield operations will live with.

The Construction Window, and Why DCA Is the Right Lesson and the Wrong One

DCA Project Journey is the cleanest available case for in-place sterile-side renovation in the MWAA system. The program completed phased reconstruction of the new sterile-area buildings between 2019 and 2021 while maintaining continuous TSA-sterile operations across the rest of the facility. Construction windows were limited to roughly 10 p.m. to 4:30 a.m. on nights when gates were not active, with a gate reactivation requirement by 4:30 a.m. regardless of construction progress. Temporary sterile partitions required daily inspection before terminal opening and periodic re-acceptance when the configuration changed. The most consequential schedule driver was not structural demolition but MEP coordination — the existing systems ran through zones that could not be shut down without affecting active gates, requiring temporary tie-ins and staged cutover sequences that added 30 to 40 percent to the MEP contractor's scope and schedule relative to greenfield equivalent work.

The chief-engineer brief estimates effective construction productivity in the occupied, sterile environment at roughly 35 to 45 percent of greenfield equivalent and flags the figure explicitly as practitioner observation, not a citable published figure. The number is the right order of magnitude for the lifecycle math, but it should not be cited as if it had a TRB report behind it. What is citable, and harder, is the night-window arithmetic. At IAD's AeroTrain station locations, the effective window narrows further because the train operates earlier in the morning and later at night than the gate operations it serves. The chief-engineer brief puts the available window at AeroTrain stations at roughly 2.5 hours per night, against 6.5 hours at DCA during Project Journey's most productive phases — again practitioner observation, not a published figure. Anyone selling a Tier 2 AeroTrain station renovation on a six-month schedule is working from equipment lead times, not from the occupied-airport construction sequence.

The TSA acceptance process compounds this. TSA does not publish a design manual for temporary sterile boundaries during construction; acceptance is governed by the local Federal Security Director, reviewing the temporary configuration against the airport's TSA-accepted security program. Under 49 CFR § 1542.107, for changes expected to last 60 days or more, the airport must file a formal proposed ASP amendment within 30 days of discovery, and TSA has 30 days under § 1542.105 to approve or deny, with requests filed at least 45 days before the proposed effective date. The 60- to 90-day project planning timeline that practitioners cite is an aggregation of these sequential steps, not a

single statutory clock — but it is real, it is critical-path, and it appears on almost no pre-construction schedule.

Vertical Circulation: Where the Geometry Fails Fastest

If the corridor is the first surface where pulse meets pedestrian, the vertical circulation is where the geometry fails fastest.

ACRP Report 67 (2012) collected more than 237,000 data points across five airports on escalator, moving walkway, and elevator performance. Its core finding for design purposes: throughput is governed by approach corridor configuration, not by installed capacity alone. The theoretical 9,000 to 12,000 passengers per hour of a high-capacity moving walkway is achievable only when approach flow is consistent and distributed across the unit width. When a single escalator at a midfield concourse station goes out of service, the vertical transition fails completely for the affected direction. Corridor flow degrades linearly with width reduction. Vertical circulation degrades binarily.

That asymmetry — linear horizontal failure versus binary vertical failure — is the highest-consequence single-point design risk in any sterile circulation system fed by an APM. It is also the failure mode most resistant to operational correction. At IAD Concourse C, the COO brief reports that one of the two escalators serving the AeroTrain platform is out of service for scheduled or unscheduled maintenance on average eight to twelve days per month. The brief attributes this to MWAA maintenance data disclosed in CIP budget justifications without specifying the CIP year, and flags that web access was unavailable during research; the figure should read as operator-chair recall pending verification against a specific MWAA CIP document. The directional claim is consistent with the practical service-life arithmetic that follows.

When an outage event happens, the working escalator carries the platform load alone, the elevator handles 12 to 15 passengers per cycle on 30-second intervals (not useful for crowd flow), and the next AeroTrain is four minutes behind. The platform does not clear in time. Dispatch extends the headway. The downstream bank push — 14 gates with a 20-minute departure window — receives passengers in a compressed wave rather than distributed across the window. Gate agents call the ops center. Facilities calls the vendor. The vendor's response time is 90 minutes minimum. The bank push is over in 40. This is documented in United's station operations logs and in MWAA's own duty manager reports, as recorded in the COO brief. The design fix — a third escalator or a second elevator bank at the Concourse C AeroTrain connection — has been estimated at \$8 to \$12 million in the COO brief, an operator-chair estimate rather than a published MWAA capital figure. The chief-engineer brief's parametric on shaft-widening for wider step-band equipment is \$3 to \$6 million per shaft pair in construction cost, exclusive of soft costs, before occupied-building premiums. The two figures are scoped differently — a third unit at the existing station versus widened shafts for replacement units — but they bracket the order of magnitude. Either way, the project has not been funded.

A service-life arithmetic the capital planning process has not yet confronted: commercial escalators and moving walkways in high-traffic airport applications have a design service

life of 25 to 30 years under ideal maintenance conditions, closer to 20 under the actual conditions of airport use, which include higher-than-rated load cycles, baggage-cart incursions, and deferred lubrication during high-tempo operations. IAD's midfield escalators, installed between 1997 and 2004, are now 22 to 29 years old. They are at or past their practical service life. In-kind replacement in an occupied sterile concourse, accounting for phasing, temporary routing, and TSA acceptance of each construction phase, runs 18 to 26 weeks per location. Any schedule showing escalator replacement in under 16 weeks is wrong.

When this end-of-life capital wave collides with capacity-driven renovation, the contractor's scope interface becomes a claims incubator: whose scope includes the deteriorated substrate under the new floor finish? Who owns the electrical service that was already at end of life before the renovation contract started? This is part of the dynamic that drives in-place renovation cost premiums above greenfield equivalents. There are two figures in the briefs for that premium, and they need to be distinguished rather than averaged. The Contrarian brief cites ACRP Report 25's general in-place renovation premium at 1.5 to 2.5 times greenfield per square foot — a published figure. The Infrastructure Economist brief estimates a 3 to 5 times multiplier for the narrower category of mid-life vertical circulation replacement at sterile-side handoffs, drawn from contractor cost indexes and pattern evidence and flagged in the brief as informed estimate rather than a verified figure. Both can be true: the 1.5-to-2.5x represents a typical in-place renovation envelope; the 3-to-5x represents the harder subset of work at TSA-sterile vertical-circulation handoffs where the construction window is shortest and the staging is tightest. Whichever multiplier you take, the capital math strongly favors investment at construction. The budget process consistently fails to capture it.

The Industry Designs to a Model It Has Never Validated

The pedestrian simulation tools that justify sterile corridor sizing — MassMotion, AnyLogic, SimWalk, Pathfinder — have been standard practice for fifteen years and longer. They produce defensible LOS projections. What they do not produce, anywhere, is a validated post-occupancy comparison.

A 2025 conference paper at the ACM Digital Library on airport simulation practice put the finding starkly: "any references reporting validation of simulation results for pedestrian behavior in airport terminals were not identified." This is the explicit conclusion of a literature review undertaken to find such references. None exist. The verification documentation that MassMotion publishes validates the model engine against controlled experimental data and published pedestrian science (Weidmann, Fruin) — validation of the simulation against the same 1970s framework the simulation was built to operationalize. It is not validation against measured airport conditions.

The sensing infrastructure to close this loop exists. Xovis reports deployment at over 120 airports and nearly 700 touchpoints; those touchpoints are queuing nodes — security, immigration, check-in — not corridor segments. Veovo's corridor-measurement capability is real and explicitly marketed for "long corridors, extended lines," but published case studies at named airports describe capability rather than documented

corridor-specific outcomes. Neither vendor has published a case study where corridor-density data changed a capital or operational decision at a named US airport.

The gap is not technical. The technology to measure corridor flow exists. The institutional practice of using that measurement to validate pre-opening design models — and to inform future corridor-sizing decisions — does not exist as documented standard at any airport in the comparator set. This is also the right answer to a sharper version of the contrarian objection that the Fruin data cuts both ways. If the underlying flow data is unreliable, the thesis cannot cite it to defend wider corridors any more than the status quo can cite it to defend current widths. The honest version is that neither side knows the actual corridor performance at IAD, that the cost of measuring is rounding error against the cost of building, and that the absence of measurement is itself the policy choice. The Tier 2 East concourse, opening September 2026, has no publicly documented corridor-level sensing deployment. If MWAA does not instrument those segments at opening, the design assumptions embedded in the simulation model will go unvalidated for the life of the building, and the next capital request for handoff capacity will have nothing more empirical to lean on than the simulation tools that produced the design no one has measured.

A counterexample within the operations-analyst brief is worth surfacing. O'Hare Terminal 2's underground pedestrian tunnel network connecting T1, T2, and T3 has adequate horizontal circulation capacity on paper. The routing — below grade, through parking infrastructure, with non-intuitive wayfinding — degrades its operational utility, and passengers default to the Skytrain even when walking would be faster. The lesson is not that corridor width does not matter. It is that the framework for evaluating sterile circulation has to include configurational variables alongside dimensional ones, and that a wider corridor placed at the wrong angle to the dominant flow is not equivalent capacity to a narrower one placed correctly. The right answer is the one this section already names: measurement, against the actual flow, in the actual building.

The Denver Confession

The cleanest available evidence that the gap between sterile corridor simulation and operating reality eventually becomes a capital problem is the Denver International Airport pedestrian walkway program announced in May 2026.

DEN will repurpose underground baggage tunnels into pedestrian walkways between Concourses A, B, and C at a cost range of \$300 million to \$700 million, with construction planned to begin 2027 and complete in 2028 or 2029. The explicit driver from DEN's own leadership: the airport has "not had redundancy for 31 years" in its inter-concourse connections. With more than 150,000 daily train riders and no pedestrian alternative, a train outage — even at 99 percent uptime, three to four full failures per year — shuts down concourse access entirely. CEO Phil Washington described the pedestrian walkways as breaking "the airport's dangerous over-reliance on aging train networks."

This is a confession.

The 1995 decision to build DEN as a single-mode APM system, without a pedestrian alternative, was not a maintenance error. It was a design choice. The argument the contrarian case would make against the 2026 thesis — that operational flexibility substitutes for physical redundancy — is the same argument that was implicitly made at DEN in 1993. That argument has cost three decades of distributed operational disruption and is now costing somewhere between \$300 million and \$700 million to correct.

The 31-year framing is not an indictment of DEN. It is a forecast for every other midfield airport that built single-mode APM access without sterile-side pedestrian redundancy, which is most of them. The IAD AeroTrain opened in 2010. The clock is running.

The Historical Arc

This is not a new debate. The aviation historian's reading is that the question has surfaced three times in the last fifty years, each time triggered by a structural break that exposed the gap between corridor design assumptions and actual operating conditions.

The first surfacing was 1973 to 1975, when the original hijacking-era security mandates revealed that terminals designed for open concourse access could not absorb checkpoint infrastructure without creating corridors that were functionally narrower post-checkpoint than pre-checkpoint. The design community responded slowly. Most airports did not invest in sterile-corridor redesign until 9/11 forced their hand nearly thirty years later.

The second was 2001 to 2004, when the Aviation and Transportation Security Act of 2001 mandated checkpoint infrastructure that consumed 1,500 to 4,000 square feet of what had been sterile corridor at most legacy terminals. The design community's response was to focus on checkpoint throughput rather than on downstream corridor geometry. TSA's design guidance addressed queuing upstream of the checkpoint but had little to say about the downstream corridor that served the enlarged concourse the checkpoint now controlled.

The third is now. The trigger is the combination of post-pandemic traffic recovery to record levels, aging vertical circulation infrastructure reaching end-of-life simultaneously, and the capital constraint that makes "design to minimum" the default. The PFC cap of \$4.50, unchanged since 2000 and reaffirmed in P.L. 118-63 (May 2024), has lost roughly 55 percent of its purchasing power against the ENR Construction Cost Index since 2000 — flagged in the historian brief as an analyst construction from primary data, against the more commonly cited 40 percent CPI figure. For a capital discussion about poured concrete and erected steel, the construction-cost baseline is the right one; that is the baseline this piece will use throughout. The headline funding instrument has lost more than half its purchasing power against the cost of the thing it funds.

Each generation built a corridor for the traffic its planners forecast and watched that corridor become the permanent operational ceiling of the next generation. Pittsburgh International's 1992 midfield terminal — one of the better-designed midfield facilities of

its generation — was sized for US Airways' hub of roughly 550 daily departures and operated at under 100 within two years after the 2002 bankruptcy restructuring. LaGuardia's Central Terminal Building, designed in 1964 for 8 million annual passengers, was processing 14 to 15 million by the 2010s — roughly 175 percent of design capacity — with sterile corridors that could not be widened without demolishing the building's load-bearing envelope. The CTB hit its operational ceiling around 11 to 12 million annual passengers and operated in degraded condition for a decade before reconstruction. JFK Terminal 1 makes the same point at the next cost increment: an \$800 million partial remediation has been weighed against a \$9 billion-plus full-replacement option, the cost trajectory that confronts a corridor problem deferred long enough that renovation no longer reaches it.

Hartsfield-Jackson is the contrast, though the comparison should rest on what is actually documented. ATL has repeatedly expanded its sterile circulation system since the 1980s and has treated corridor width as a multi-decade design variable rather than a residual after gate count and checkpoint area were fixed. The plane-train corridors below the apron were sized with two-direction flow as a design premise, not as an afterthought, and the Concourse T addition was drawn with wider spines than the original A-through-E generation. ATL's sterile circulation has aged better than most comparable hubs. The lesson is not that Atlanta had better designers. It is that Atlanta's management treated corridor width as a multi-decade capital investment, not a current-cycle cost to minimize.

The IAD case is on the LGA arc, not the ATL arc. The midfield concourses were designed to a traffic model calibrated on late-1990s United hub operations. The AeroTrain was retrofit onto an existing sterile envelope in 2010, meaning the handoff geometry was not designed as a whole — it was made to fit a building that was already a decade old. United's hub has restructured twice since the original design year. The corridor geometry has not.

Why the Capital Decision Loses Every Internal Competition

The reason the capital decision keeps losing — at IAD, at every midfield hub, in every CIP cycle — is not that it is wrong. It is that the accountability structure is wrong.

Sterile circulation is common-use infrastructure that produces no revenue. Every capital allocation framework a large hub operates under — residual rate protection, signatory airline Majority-in-Interest thresholds, PFC eligibility screens — structurally disadvantages spending that cannot be tied to a gate, a holdroom, or a check-in position. Rating agencies do not have a line item for corridor resilience. They do have line items for operating disruptions, litigation exposure, and single-point-of-failure infrastructure risk. The translation from "this corridor will jam at peak someday" to "this is a credit event in the making" is not automatic. The CEO has to make it, explicitly, before the event.

The airline commercial reality compounds this. United Airlines is the dominant carrier at IAD; the airline-strategist brief reads United's enplanement share at 60 to 65 percent

while the CEO brief reads roughly 50 to 55 percent of seats. Both figures should be verified against current DOT T-100 data and the most recent MWAA reporting. At any value in that range, United triggers Majority-in-Interest review on material common-use capital. The Majority-in-Interest clause in MWAA's airline use-and-lease agreement gives United structural veto power over capital projects above defined thresholds. A corridor recommissioning project that adds material cost to the common-use rate base — even if physically located in a concourse United dominates — will require United's tacit or explicit approval to proceed without rate-setting conflict.

The misalignment runs deeper. United's preferential gates at the midfield drive the peak loads in the corridors. The corridors themselves are common-use. United pays for corridor capacity as a common-use cost at its enplanement share, not as a preferential cost reflecting its load contribution. This gives United limited incentive to advocate for corridor upgrades and strong incentive to characterize the problem as an airport operations issue rather than a shared capital problem. United's posture is self-serving and also not entirely wrong — some corridor performance problems at IAD are operational rather than dimensional. The airport's task is to distinguish which problems are which, and to make the case in terms United's internal teams will credit.

The case that will move United's network planning team — and the airline-strategist brief is unambiguous that this is the team that matters, not airline relations — is the one expressed in connect-bank reliability and minimum-connection-time terms. Delta's internal justification for the 2012-2014 Atlanta T-concourse redesign included connect-time reduction as a quantifiable benefit measurable in revenue terms: a one-minute reduction in effective connect time across a major bank can open or close hundreds of itinerary combinations in the carrier's inventory system. When Denver International undertook its South Terminal Redevelopment Program in 2016 to 2021, the corridor and connector elements were structured as common-use improvements with United's explicit buy-in achieved through a combination of CPE cap commitments and connecting-time analysis that United's network planning team accepted. The airport did not present it as a resilience investment. It presented it as a hub efficiency investment. The distinction mattered. It is the only frame in which an MWAA corridor capital request survives a United MII review.

The political economy at MWAA closes the trap. The 2012 DOT OIG audit found that MWAA awarded 117 of 190 contracts exceeding \$200,000 — 62 percent — using categorical exceptions that limited competition, totaling \$225 million or 40 percent of total contract value. The Silver Line Phase 2 added more than \$200 million in contractor cost overruns settled in 2022. The institutional memory of congressional hearings and OIG audits makes MWAA leadership cautious about capital programs that are difficult to explain and easy to second-guess. Invisible infrastructure — a wider corridor, a redundant escalator bank — is exactly the kind of capital that gets cut when board members are anticipating the next OIG call.

The PFC cap and the BIL/IIJA Airport Terminal Program complete the picture. The \$4.50 cap is not moving; airlines' structural advantage in the PFC debate at the

Congressional level is durable. BIL/IIJA ATP grants — \$1 billion annually for FY2022 through FY2026, 80 percent federal share at large hubs — are the available non-airline funding lever. They have been oversubscribed at every cycle. Projects that advance have visible outputs: new checkpoint buildings, new concourses, ADA upgrades. DCA Project Journey received \$269.8 million in ATP funding against its \$1 billion overall program — but it was a checkpoint replacement program. A sterile corridor hardening project without a headline feature will struggle in this competition unless it is bundled into a larger program scope.

The Lifecycle Accountability Gap

Underneath all of this — beneath the LOS framework, the column grid, the airline veto, the OIG memory — is the structural pathology the operator's intuition pass surfaced most clearly.

Every variable that determines whether a sterile corridor works is set by someone who will be retired or in a different job before the first passenger walks through it. The person who signs off on 40 feet of clear corridor width instead of 50 — because the drawing fit better and they were three weeks from their deadline — will never stand in that corridor at 7 a.m. on a Tuesday in February when the AeroTrain is delayed 12 minutes and four United banks are arriving simultaneously. The lifecycle accountability gap is the sterile corridor problem. Everything else is downstream of it.

This is the same class of mistake the industry made with jet-bridge specifications in the late 1970s. Designers locked the bridge geometry to the aircraft mix of the day — B-727, DC-9, early 737. Then the B-757 came in with a different door sill height. Then the 767. Then the A330. Then the 777. At virtually every airport built before 1990, there are gates that handle certain aircraft but not others because the bridge spec was locked to a fleet two technology generations ago. The workaround became permanent. The sterile corridor width decision is the same mistake made for the same structural reason: the person making the decision is optimizing for current-cycle cost and is not in the room for the operational consequence.

The stadium industry's response to the analogous problem illustrates what a different accountability structure looks like. The generation of NFL stadiums built between 2014 and 2022 — Allegiant in Las Vegas, SoFi in Inglewood, Mercedes-Benz in Atlanta — used measured event-day flow from the previous generation as design input, not as soft reference. The practice has been described in the premium segment of the industry as a contractual mechanism: the team's operations staff signs off on flow before certificate of occupancy. If the measured event-day flow does not meet the spec, there is a remedy. Aviation has no equivalent. The planning manual says "consult LOS standards." There is no contractual mechanism that ties the designer's fee to measured throughput at year-five occupancy. This is the gap that produces corridors that work fine at opening and fail at year seven.

IV. The Counter-Case, Honestly Presented

There is a serious version of the argument against this thesis, made by sophisticated planners and operators inside the industry. Several of its claims survive the empirical record better than the thesis would prefer.

The strongest objection is the path-dependence one, because it goes after the thesis's central rhetorical claim and partially wins.

The thesis says design decisions are locked in for ten years or more, and that the geometry committed at IAD now will govern operations for thirty. The contrarian brief points out that this is true of greenfield construction but substantially less true of the airports the thesis actually discusses. DCA's Project Journey is the thesis's own example of in-place sterile-side renovation — and it proves the opposite of what the thesis implies. The concourse rebuilds at DCA, phased through 2019-2021, reconfigured sterile corridor geometry, replaced vertical circulation, and widened circulation paths in an occupied, operating concourse. The project was completed without a single day of full-concourse closure. It was operationally complex and expensive, but it was done. IAD's own Tier 2 midfield renovation program is an ongoing demonstration that corridor geometry at IAD is not locked in; MWAA is actively rebuilding those design variables right now. To argue simultaneously that corridor decisions are locked in for decades and that the current Tier 2 program is the window to fix them is to claim, in the same sentence, that the window is closed and the window is open.

The contrarian's strongest version is that the renovation window is not narrow at all — it opens continuously as concourses cycle through major renovations, and the right framing is not "act now or lose for thirty years" but "use this cycle, and the next one, and the one after that." On its strongest version, this collapses the urgency the thesis claims.

The cost-benefit objection is the next-strongest, and it does real damage to the throughput framing. The contrarian brief argues that the corridor segments at IAD that are genuinely constrained at peak are primarily at the AeroTrain station heads, not along the full concourse spines. The number of hours per year that those segments operate above LOS D conditions is a critical input the thesis does not supply. If that number is, say, 200 hours per year — less than 2.5 percent of operating hours — the capital case for widening is substantially weaker than a narrative built around vivid peak-period observations suggests. ACRP Report 25 documents that in-place renovation of occupied concourses typically costs 1.5 to 2.5 times the equivalent greenfield construction cost per square foot, before TSA coordination premium. The relevant counterfactual is not wide corridor versus narrow corridor. It is wide corridor versus better operations in the narrow corridor.

The third objection is operational, and it is correct in the middle of the failure distribution. A cleaning schedule that avoids peak periods is free; a wider corridor that can accommodate a cleaning crew without reducing effective width below LOS C is not. ACRP Report 83 documents that escalator mean-time-to-repair at US airports averages 4 to 6 hours without priority maintenance contracts and falls to under 90 minutes with

them. A maintenance SLA requiring restoration within 45 minutes during peak periods, enforced with financial penalties, costs less than a redundant escalator bay and addresses the same failure mode. The contrarian's harder point is the prior-question version: until MWAA can show that its operating budget for sterile-side corridor operations has been optimized — cleaning schedules, maintenance SLAs, incident response staffing, AeroTrain headway management — the capital case is asking the rate base to fund what the operating budget has not been pressed to do.

The checkpoint-trade-off objection is the one the run prompt specifically excluded from scope, and the contrarian brief is right to push back on the exclusion. At IAD, the actual constraint on passenger throughput on a disrupted afternoon is the number of lanes open at the midfield security checkpoints, TSA staffing, and hold-room capacity at gates accepting diverted passengers. These are upstream and downstream of the sterile corridors the thesis focuses on, and they are the binding constraints under disruption in the middle of the failure distribution. If MWAA has \$50 million in discretionary resilience capital — roughly what a meaningful sterile-corridor widening program at one IAD concourse would cost — the expected throughput gain from deploying that capital at checkpoints almost certainly exceeds the throughput gain from wider post-checkpoint corridors at the median. The thesis owes that comparison a direct answer rather than a categorical dodge.

The empirical objection is the cleanest. The thesis calls for an empirical treatment of width-to-throughput curves cited to actual measurement, not LOS letter grades. The empirical record on airport-specific pedestrian flow is thinner than the thesis would prefer. If the Fruin data is unreliable, the thesis cannot cite it to support specific corridor-widening targets any more than the status quo can cite it to defend current standards. The data uncertainty is an argument for better instrumentation before capital decisions, not a foregone conclusion that capital is the answer.

The political objection is the structural one and it does not weaken with rebuttal. The PFC cap at \$4.50 is not moving. The \$22 billion IAD revitalization proposal floated by the Trump administration DOT in December 2025 has no confirmed federal funding mechanism as of May 2026; DOT did not respond to press requests for comment on the proposal. When a \$22 billion program is the headline conversation, everything else — including the sterile corridor resilience investments the thesis advocates — is subordinated. A planning process waiting on DOT approval for a \$22 billion bond program cannot simultaneously prioritize incremental investments in corridor redundancy. The CEO who picks a fight with United over corridor width is spending political capital that might be needed for the next holdroom negotiation, the next gate reallocation, or the next federal policy fight where United's advocacy matters.

These are not weak objections. The path-dependence one, in particular, partially succeeds and forces the thesis to give back rhetorical ground. The remainder of the piece concedes what must be conceded and then argues that the thesis still holds — for narrower reasons, in a more bounded form, than the v1 of this argument might have suggested.

V. Why the Counter-Case Is Insufficient

DCA Project Journey did what the contrarian brief says it did. It reconfigured sterile corridor geometry, replaced vertical circulation, and widened circulation paths in an occupied, operating concourse. The framing that "design variables locked in 10+ years ago determine outcomes" is too strong if it implies the corridors are physically immutable. They are not. MWAA is rebuilding them right now. The contrarian is correct that the rhetorical force of "act now or lose for thirty years" depends on a path-dependence claim that the in-place renovation record partially contradicts.

What survives is a narrower claim about the bounded character of in-place renovation. The renovation that DCA achieved happened inside an inherited structural envelope. It moved partition walls, replaced equipment, and reorganized circulation against the existing column grid. It did not relocate the columns. It did not change the structural bay spacing. It did not expand the building footprint at the gate side. What in-place renovation can change is the configuration inside the envelope. What it cannot change, at any cost the rate base will tolerate, is the envelope itself. That envelope is the binding constraint on a 1990s midfield. It is the thing the Tier 2 program is setting for the 2030s and 2040s.

The mid-life cost multiplier sharpens the practical version of this claim. For general in-place renovation, ACRP Report 25's 1.5 to 2.5 times greenfield multiplier is the documented industry figure and the right number for most of the work DCA Project Journey was doing. For the narrower category of mid-life vertical circulation replacement at sterile-side handoffs — the work this thesis specifically cares about — the Infrastructure Economist brief's 3 to 5 times multiplier is the right order of magnitude, flagged in that brief as an informed estimate rather than a published figure. Both numbers describe real categories of work; the higher multiplier applies where the construction window is shortest (the 2.5-hour AeroTrain station window), the staging tightest, and the TSA acceptance cycle most consequential. At those nodes, the practical window does not reopen continuously. It reopens once per generation, because the cost of opening it more often exceeds what the political and financial system will tolerate. JFK Terminal 1 sits at the end of that arc: an \$800 million partial remediation against a \$9 billion-plus full replacement, the increment that becomes available when the renovation window has been allowed to close. The thirty-year framing is a financial claim about how often the rate base will absorb the multiplier, not a structural claim that the concrete cannot be touched. That softer version of path-dependence is what survives the contrarian's challenge, and it is enough.

The cost-benefit objection forces a similar narrowing. The contrarian brief's question — how many hours per year do IAD corridor segments actually operate above LOS D — is the right question, and the honest answer is that nobody knows because nobody has measured. The 200-hour speculation could be true. It could be a tenth of that, or three times that, and the design conversation cannot tell you which. The instrumentation recommendation that closes Section III is the only honest response: measure first, with

sensors the cost of which is rounding error against the capital scope of the building going up, and let the measurement decide which corridor segments justify the capital and which do not. The thesis's claim is not that every IAD corridor is undersized. It is that the existing design framework cannot distinguish between segments that are and segments that are not, and that the right precondition for a capital decision is to install the sensing that produces a defensible measurement.

The operations objection is correct in the middle of the failure distribution and wrong at the tail, and the thesis lives in the tail. The Contrarian is right that MWAA almost certainly has not optimized cleaning schedules, maintenance SLAs, incident response staffing, and AeroTrain headway management to the extent its operating budget permits. The piece should concede that and concede that the operational stack should be funded in parallel with any capital scope — explicitly so. Resisting the operational work would weaken the capital case rather than strengthen it. But operational interventions move the median performance of the system. They do not compress the tail. When the only escalator at a sterile transition point is out of service for 90 minutes and there is no second unit, no operational SLA recovers that 90 minutes.

The cleanest version of the tail-failure case is the one the virtual-christian brief surfaces and the v1 of this draft missed. What happens at IAD when the AeroTrain goes down for four hours, not four minutes? MWAA has a plan. The plan is buses. The buses use the airfield roads. The sterile integrity of the bus-to-concourse transition is technically fine because there are procedures for it. But the throughput is not fine. The bus alternative is not designed for four-hour duration; it is designed to bridge the gap until AeroTrain comes back. If it does not come back for four hours — weather, mechanical, propulsion-system event, fire alarm cascade — the corridor capacity question becomes almost irrelevant because the upstream delivery system is broken. The real resilience question at IAD is not "can the corridors handle peak flow." It is "what does the system look like when the AeroTrain is not running and we need to move eight thousand passengers across the airfield by bus and still maintain sterile integrity." That scenario lives almost nowhere in the published MWAA capital documents and was not, to the operator's recollection, modeled in the original AeroTrain design case. The DEN parallel is most pointed here. Denver's \$300-to-\$700-million pedestrian alternative is being built precisely to address the four-hour-failure case, not the daily-degradation case. The single-mode APM is the problem the DEN program is buying its way out of. IAD has not yet bought.

The checkpoint trade-off objection deserves a direct quantitative answer, not a category dodge. The thesis should concede the comparison the contrarian brief proposes and reframe the conclusion. Checkpoint capital and corridor capital are not substitutes; they live in different parts of the failure distribution and address different failure modes.

Checkpoint capital — additional CT lanes, additional staffing during disruption windows, supplementary screening capability — moves the median of the failure distribution. On a Tuesday afternoon when the checkpoint backs up, more lanes shorten the wait. The throughput-gain-per-dollar at the checkpoint is genuinely large in the middle of the

distribution because the median failure mode at the checkpoint is queueing under load, and queueing under load responds to capacity addition. The contrarian is right that \$50 million deployed at IAD checkpoints would produce a measurable throughput improvement that \$50 million deployed at corridor widening probably would not.

The tail of the distribution is the asymmetric part. At the median, checkpoint capacity is the binding constraint and corridor capacity is not. At the tail — the eight to ten percent of days the COO chair identifies as marginal, the two-or-three days a year that escalate to crisis — the binding constraint shifts. When the AeroTrain fails for four hours, no checkpoint capacity recovers it. When the single working escalator at the C station fails on a snowy Tuesday morning, no checkpoint staffing addresses the binary loss of vertical circulation at the APM handoff. The throughput-gain-per-dollar comparison the contrarian proposes is correct on the median dimension and wrong on the tail dimension, because the failure modes are categorically different. A checkpoint that operates at degraded capacity is still operating. A vertical circulation node with no working unit and no redundant unit is not operating.

The right framework is not to choose between checkpoint capital and corridor capital. It is to recognize that they sit in different parts of the capital stack against different failure-mode classes, and to fund both — the checkpoint capital to compress the median, the corridor and vertical circulation capital to compress the tail. The contrarian's expected-value framework is the right one to use; it just produces a different conclusion than the contrarian intends when the value distribution it operates over is correctly specified. The thesis is making a tail-compression argument. The checkpoint argument is a median-improvement argument. Both are valid. Neither substitutes for the other.

On the political moment. The \$22 billion proposal is real, and the gravitational pull of a transformational announcement is real. The CEO's calculus about not picking the wrong fight with United is also real. What these arguments concede is that the political environment is unfavorable. They do not address whether the underlying physics of the corridor changes because the politics are unfavorable. It does not.

The framing question is not whether to fight the political environment but how to structure the capital request so it survives the political environment. The DEN South Terminal Redevelopment precedent shows what works: a corridor capital program structured with explicit connecting-time analysis accepted by United network planning, presented as a hub efficiency investment rather than a resilience investment, with CPE cap commitments to manage the MII process. The DCA Project Journey precedent shows what is fundable within the federal grant stack: a corridor program bundled inside a larger headline project. A corridor-resilience capital request, framed as resilience, presented in isolation, in the middle of a \$22 billion conversation, will not survive United's MII review, will not win an ATP grant, and will not move at MWAA's board. The framing must change. The funding stack must be assembled. The connection between corridor performance and United's connecting-bank reliability must be measured and demonstrated. The Tier 2 program must be the vehicle.

The honest version of the thesis after the counter-case is narrower and harder than the version that opened this piece. Path dependence is not absolute; the renovation window is real and recurring, but the financial multiplier on mid-life work at TSA-sterile vertical-circulation handoffs makes the practical window once-per-generation rather than continuous. Operational interventions move the median; only redundant infrastructure compresses the tail, and the tail at IAD is the four-hour AeroTrain failure scenario that no operational fix addresses. Checkpoint capital and corridor capital are not substitutes. The political moment is hostile; the framing must change, the funding stack must be assembled accordingly, and the corridor scope must be bundled, not freestanding.

This is what survives the counter-case. It is enough. It is also less than the v1 of this argument was claiming, and the discipline of saying so out loud is the difference between an argument and a pitch.

VI. Implications for MWAA

It is 6:47 a.m. now. The cleaning crew has cleared the south end of the corridor. The rough escalator has not tripped. The 0700 bank is going to make it, the way 285 mornings a year make it, the way the operator's twenty-five years of experience tells him most marginal mornings make it. The next bank arrives at 7:39. The next escalator inspection is overnight.

The decisions that determine whether the 7:39 bank ten years from now arrives into a corridor that absorbs it or one that fails it are being made now, in conference rooms on Saarinen Circle, at 30 percent design development on the next Tier 2 phase, in the use-and-lease conversations that have not yet been scheduled. The 6:45 a.m. scene at the start of this piece is not the failure case. It is the baseline. The thesis is about what the baseline looks like in 2036, when the escalators installed in 2002 are 34 years old, the column grid is the column grid, and the international-to-international flow that the original midfield was not designed for has become the dominant connecting case at C and D.

The single most useful thing MWAA can do in the next eighteen months is also the cheapest. Instrument the AeroTrain handoff points at Tier 2 East from the day it opens in September 2026. Xovis 3D stereo-optical sensors at the vertical circulation nodes. Veovo or BlipTrack flow tracking through the connector corridor segments. LiDAR for the larger volumes if the pilot data justifies it. The capital cost is small enough to fit inside a state-of-good-repair line item — likely low single-digit millions for a single concourse's vertical handoffs and connector segments, against a concourse program in the hundreds of millions; this is analyst sizing rather than a vendor quote, and a competitively-bid scope would set the actual number. The value is not real-time operational management, though that is real. The value is in the first published post-occupancy throughput baseline for an AeroTrain handoff at IAD, against the simulation predictions that justified the design. If MWAA does not do this at opening, the design assumptions go unvalidated for the life of the building, and the next capital request for

handoff capacity has no empirical foundation other than the same simulation tools that produced the design no one has measured. The instrumentation also addresses the contrarian's prior-question objection — that operational optimization has not been exhausted — by producing the data against which both operational and capital interventions can be evaluated.

The Concourse C escalator fix is next, because it is the cheapest legible capital intervention against the documented failure mode. The COO brief's \$8 to \$12 million range for a third escalator or a second elevator bank is operator-chair memory rather than a published MWAA capital estimate; the chief-engineer brief's \$3 to \$6 million per shaft pair for widened-step-band replacement is differently scoped but brackets the order of magnitude. Either figure is small enough to fit inside a state-of-good-repair allocation rather than requiring a discrete MII vote. The historical narrative ready to be delivered to the board and to the rating agencies is that the AeroTrain capital history of the 2000s produced a lesson about vertical circulation sizing at handoff points, and this is the proactive correction of that lesson before it becomes a credit event. The agencies understand that argument; they are less receptive to "this corridor gets congested at peak."

The use-and-lease window is the strategic moment to restructure how corridor capital is classified. MWAA's posture should not be confrontational. It should be analytical. Present United with the corridor-load data that the Tier 2 East instrumentation begins to generate. Show the misalignment between United's enplanement share and its peak-corridor-load share at the AeroTrain handoff nodes. Propose a structure in which corridor capital that follows preferential gate clusters is treated as preferential infrastructure rather than common-use. This is the only structural change that aligns the carrier whose banking operation drives the demand with the cost of meeting that demand, and it is the conversation that requires data to be possible. The instrumentation makes the conversation possible.

The case that survives the United MII review is the connect-time case, not the resilience case, and the airline-strategist brief is unambiguous about where it needs to be made. The audience is not airline relations. The audience is network planning. MWAA's tactical sequence: instrument the Tier 2 East handoffs at opening; spend the first six months of data accumulation establishing the corridor-performance-to-connect-bank-reliability link; build a connect-time-reduction case structured the way Delta structured the 2012-2014 Atlanta T-concourse redesign; present that case to United network planning with the same CPE cap commitments DEN used in the 2016-2021 South Terminal Redevelopment Program; only then bring it to airline relations as a use-and-lease conversation. Hub efficiency investment, not resilience investment, is the framing that produced United's buy-in at DEN. It is the framing that survives.

On the \$22 billion proposal. The political logic that subordinates everything to the headline program is real, and a freestanding corridor resilience investment will not survive. MWAA should advocate explicitly for the corridor resilience scope to be bundled into the \$22 billion program at the planning stage. The regulatory-political brief reports

the concept as including a \$3.75 billion underground central spine element that would expand AeroTrain connectivity and render the mobile lounges obsolete — that is the program as floated, not the program as recommended. MWAA should propose that scope be expanded to include not just additional AeroTrain capacity but expanded vertical circulation at all handoff points and parallel sterile-side pedestrian paths between concourses as DEN is now building. If the \$22 billion program proceeds, the corridor resilience scope is funded inside the larger stack. If it does not, MWAA falls back to the Tier 2 program with the Concourse C escalator addition as the discrete intervention. Either way, the corridor scope is positioned, not deferred.

The federal funding stack rewards visible outputs; resilience scope wins inside checkpoint or concourse projects, not as a standalone. MWAA's next ATP application should bundle corridor scope into the headline project of that cycle — Tier 2 phases, the AeroTrain expansion if it advances, or a follow-on to Project Journey at DCA. The PFC cap will not move on a project timeline; the assumption is that PFC revenue continues to back bond issuance for the broader capital program and that corridor scope must compete inside that envelope.

The international-to-international flow deserves to be a separately modeled design case in the Tier 2 program, not a footnote to the domestic flow modeling. IAD's international traffic mix is unusually high for a US airport its size, and the sterile re-entry flow for connecting international passengers — arriving on an international flight, clearing CBP, re-entering the sterile zone, catching a connection — is a distinct use case with different spatial requirements, different dwell patterns, different time-pressure distributions. The Tier 2 East concourse includes a sterile mezzanine corridor for international inbound passengers and direct AeroTrain connection. If that corridor is designed for the 7 a.m. domestic bank and turns out to be the wrong design for the noon international connection complex, MWAA has optimized for the metric it could measure rather than the flow that will actually strain the system. The instrumentation recommendation pays double here: the same sensors that produce the connect-time data for United also produce the international-to-international flow data that lets MWAA evaluate whether the noon complex requires geometry the domestic case did not.

The operations work belongs alongside all of this and the thesis is better with it than against it. The cleaning window at Concourse C is a contract problem, not a capital problem; consolidate corridor-performance accountability under a single operations leader who has authority over both the cleaning contract and the corridor flow targets, and revise the contract on the next renewal cycle to align the window to the United bank structure. The escalator maintenance SLA is a contract problem; the 45-minute peak-period restoration window that ACRP Report 83 documents as achievable with priority contracts is the right specification, and the cost of that contract is materially below the cost of the third unit it cannot substitute for. The AeroTrain headway management at the Concourse A/B platform — currently maintained by an operations representative with manual hold authority — should be folded into the consolidated operations scope rather than papered over indefinitely. None of this work substitutes for the capital case. It

strengthens it by demonstrating that the operating budget has been pressed to its limit before the capital case is made.

It is 6:51 now. The operator is on his second coffee. He is thinking, as he watches the rough escalator carry the last of the bank push, about the 80 mornings a year that are not this one — the ones where the rough escalator does trip, or two of three checkpoint lanes are out, or the cleaning crew has not cleared and the bank arrives early. He has been making the case for that math for years, in language the board has not quite found a way to act on. The number he wishes someone would build into the CIP narrative is not the average corridor LOS; it is the share of mornings in the lower tail of the distribution, and the share of that tail that a redundant escalator or a wider exit throat would have moved into the meat of the curve. That is the metric the rating agencies would understand. It is the metric the board's audit committee would understand. It is the metric the operator has carried in his head for years, and the one the capital program has not yet learned to write down.

A corridor is a piece of floor between two ceilings. It is the cheapest element to draw, the easiest to under-size, and the most expensive to fix. The Tier 2 East concourse opens in September 2026 with corridor geometry that will be the corridor geometry of IAD for thirty years. The decision MWAA has not yet made — whether to instrument that geometry on opening day, against the simulations that justified it — is the smallest, cheapest, and most consequential decision left in the program. The 7:39 bank ten years from now is the one that should be on the board's mind. The 6:45 a.m. scene is the one already on the operator's.

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